Monte Carlo Simulation of Ballistic Particles Bouncing and Sticking on a Planar Surface

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(Dated: October 22, 2026)

We present a Monte Carlo simulation of non-interacting particles emitted from a point source on a planar floor, moving under gravity and linear drag, and undergoing stochastic sticking, specular reflection, or diffuse reflection at each impact. Initial speeds are drawn from a Maxwell–Boltzmann distribution by rejection sampling and emission directions from a Lambertian (cosine-law) distribution by inverse-transform sampling. Trajectories are integrated with an embedded Runge–Kutta (Dormand–Prince) method, and floor crossings are resolved by linear interpolation within the time step. The implementation is validated with thirteen checks against analytically known results, including the ballistic range, a three-dimensional random walk, the geometric distribution of bounce numbers, and the terminal velocity. We show analytically and confirm numerically that the dominant discretization error arises from the landing interpolation and scales as Δt^2, with mean energy loss $mg^2\Delta t^2/12$ and mean range error $\pi g\Delta t^2/24$ per landing; the fitted order is 2.00. At $\Delta t = 0.01$ this error is more than three orders of magnitude below the statistical uncertainty for $N = 10^4$ particles, so the accuracy is sampling-limited. The computational cost scales as $O(N/(p_{\mathrm{stick}}\,\Delta t))$, in agreement with measured particle-step counts to within 1% for $N \geq 10^4$. Specular reflection yields a mean final distance about 2.7 times larger than diffuse reflection, and drag reduces the mean final distance by a factor of 2.6–4.7.

I. INTRODUCTION

The transport of particles that are repeatedly scattered or adsorbed at a surface is a recurring problem in physics and engineering, appearing in granular ballistics, droplet and aerosol deposition, and rarefied-gas flow in vacuum systems, where diffuse wall scattering is modeled by the Knudsen cosine law [1]. A common feature of these systems is that each particle performs a sequence of free flights separated by stochastic surface interactions, and the observable of interest is the distribution of final positions after the particle is captured.

In this work we study a minimal model of this class: particles emitted from a point source on a single plane, moving under gravity and linear drag, and either sticking or reflecting at each impact. The model is deliberately simple. Because there is a single surface and the free-flight equation of motion is linear, many observables—the single-flight range, the bounce-number distribution, the mean-squared displacement under diffuse scattering, and the terminal velocity—are available in closed form. This makes the model well suited to a systematic study of the numerical method itself: we can separate and quantify round-off, integration, discretization, and sampling errors, and compare the measured computational cost with an analytic estimate.

The paper is organized as follows. Section II defines the model, and Sec. III the sampling of initial conditions. Section IV describes the numerical method. Section V presents the validation against analytic results, Sec. VI the error analysis, and Sec. VII the computational cost. Physical results are given in Sec. VIII, and Sec. IX summarizes our findings and the limitations of the model.

II. MODEL

A. Geometry and units

The floor is the plane $z = 0$, with gravity along $-\hat{z}$. All particles start at the origin and do not interact with one another. We use units with $m = g = 1$; with the thermal velocity scale σ introduced in Sec. III, lengths are measured in units of σ^2/g and times in units of σ/g. Unless stated otherwise, $\sigma = 1$.

B. Equations of motion

Between impacts each particle obeys

$$m\,\ddot{\vec{r}} = -mg\,\hat{z} - \gamma\,\dot{\vec{r}}, \tag{1}$$

where γ is the linear drag coefficient. Written as a first-order system with $\beta \equiv \gamma/m$,

$$\dot{\vec{r}} = \vec{v}, \qquad \dot{\vec{v}} = -g\,\hat{z} - \beta\,\vec{v}. \tag{2}$$

Equation (2) is linear and has the exact solution

$$\vec{v}(t) = -\vec{v}_t + (\vec{v}_0 + \vec{v}_t)\,e^{-\beta t}, \tag{3}$$

$$\vec{r}(t) = \vec{r}_0 - \vec{v}_t\,t + (\vec{v}_0 + \vec{v}_t)\,\frac{1 - e^{-\beta t}}{\beta}, \tag{4}$$

with $\vec{v}_t = (g/\beta)\,\hat{z}$, so that a falling particle approaches the terminal speed $|\vec{v}_t| = mg/\gamma$. Equations (3) and (4) serve as a reference solution for the integrator. In the drag-free limit, a particle launched with speed v at polar angle θ (measured from the vertical) has flight time $2v\cos\theta/g$ and range

$$R = \frac{v^2 \sin 2\theta}{g}. \tag{5}$$

C. Surface interaction

At each impact the particle sticks with probability p_{stick} and is removed from the dynamics. Otherwise it is reflected, either *specularly* with probability p_s ($v_z \to -v_z$, tangential components unchanged), modeling a smooth surface, or *diffusely* with probability $1-p_s$, in which case the speed is preserved and a new direction is drawn from the cosine law of Sec. III B, modeling a rough surface. All reflections are elastic, so in the absence of drag the kinetic plus potential energy is conserved between capture events.

Since successive impacts are independent Bernoulli trials, the number of bounces k before capture is geometrically distributed,

$$P(k) = (1-p_{\mathrm{stick}})^k\, p_{\mathrm{stick}}, \qquad \langle k\rangle = \frac{1-p_{\mathrm{stick}}}{p_{\mathrm{stick}}}, \tag{6}$$

and each particle performs $k+1$ flights.

III. SAMPLING OF INITIAL CONDITIONS

A. Speeds: rejection sampling

For systems with a large number of particles it is impractical to follow the speed of every individual particle. Instead, we model the distribution of speeds on a continuum by a *probability density function* (PDF) $f(v)$, which gives the relative likelihood of finding a particle at speed v. The probability that a particle has a speed of at most v is given by the *cumulative distribution function* (CDF)

$$\mathcal{P}(v) = \int_0^v f(v')\, dv', \tag{7}$$

where the PDF is normalized such that $\mathcal{P}(\infty) = 1$.

Assuming that the emitted particles have a common mass m and are in thermal equilibrium at temperature T, their speeds follow the Maxwell–Boltzmann distribution

$$f(v) = \sqrt{\frac{2}{\pi}}\, v^2 \left(\frac{m}{k_B T}\right)^{3/2} e^{-mv^2/(2k_B T)}, \tag{8}$$

where k_B is the Boltzmann constant. With the thermal velocity scale $\sigma = \sqrt{k_B T/m}$ this becomes

$$f(v) = \sqrt{\frac{2}{\pi}}\, \frac{v^2}{\sigma^3}\, e^{-v^2/2\sigma^2}. \tag{9}$$

The integral in Eq. (7) has no elementary closed form for Eq. (9), so the CDF cannot be inverted analytically. We therefore use von Neumann rejection sampling [2, 3], which requires only the PDF itself. Points (v, y) are drawn uniformly in the rectangle $[0, v_{\max}] \times [0, f_{\max}]$, and v is accepted if $y \le f(v)$ and rejected otherwise. The accepted points are uniformly distributed over the area under the curve, so their abscissae v are distributed according to $f(v)$.

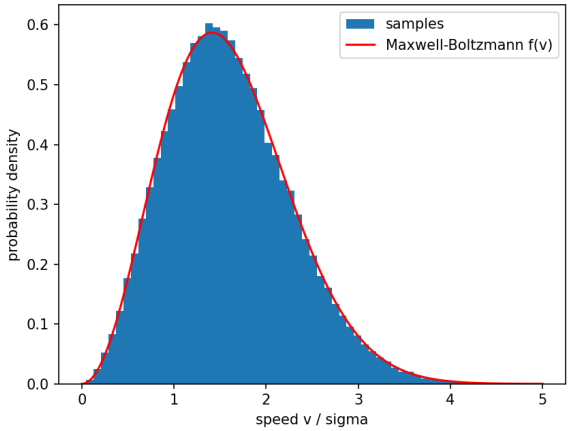


FIG. 1. Histogram of 10^5 sampled speeds compared with the Maxwell–Boltzmann distribution, Eq. (9).

The height of the rectangle is the maximum of the PDF. Setting $df/dv = 0$ gives the most probable speed $v_{\mathrm{peak}} = \sqrt{2}\,\sigma$ and $f_{\max} = f(v_{\mathrm{peak}}) = \sqrt{8/(\pi e^2)}\,/\sigma$. For the width we choose $v_{\max} = 5\sigma$, where the PDF is already negligible: the truncation discards a fraction 1.5×10^{-5} of the distribution, which is negligible at the sample sizes used here.

The expected acceptance rate is the ratio of the area under the curve to the area of the rectangle, $1/(5\sigma f_{\max}) = \sqrt{\pi/2}\, e/10 = 0.341$; we observe 0.340 for 10^5 samples. The sample mean speed is $1.594\,\sigma$ (exact: $2\sqrt{2/\pi}\,\sigma = 1.596\,\sigma$) and the sample mean of v^2 is $2.99\,\sigma^2$ (exact: $3\sigma^2$). The histogram is shown in Fig. 1.

B. Directions: inverse-transform sampling

The rejection step yields only the magnitude of the velocity; its direction is described by the polar angle θ, measured from the vertical, and the azimuth ϕ (physics convention). In spherical coordinates the solid angle element is

$$d\Omega = \sin\theta\, d\theta\, d\phi. \tag{10}$$

Particles leaving a surface follow the Lambertian cosine law: the number of particles per unit solid angle is proportional to $\cos\theta$. Combined with Eq. (10), the polar-angle distribution is $p(\theta) \propto \cos\theta \sin\theta$ on $[0, \pi/2]$, which after normalization reads $p(\theta) = \sin 2\theta$ with CDF $F(\theta) = \sin^2\theta$. The emission is rotationally symmetric about the vertical, so ϕ is uniform on $[0, 2\pi)$ with CDF $\phi/2\pi$. Setting each CDF equal to a uniform deviate $u, u' \in [0, 1)$ and inverting gives

$$\theta = \arcsin\sqrt{u}, \qquad \phi = 2\pi u'. \tag{11}$$

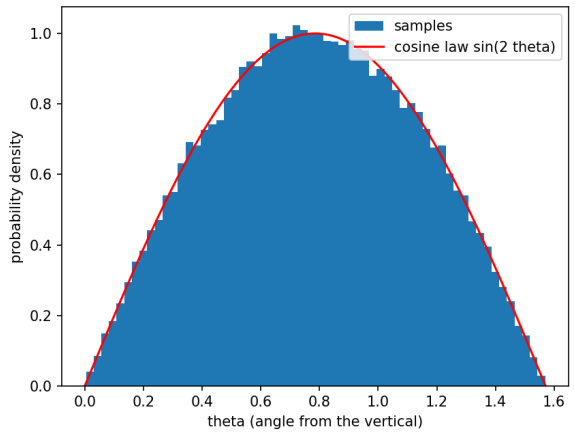


FIG. 2. Histogram of 10^5 sampled polar angles compared with $p(\theta) = \sin 2\theta$.

Algorithm 1 Particle emission, flight, and surface interaction.

Place N particles at the origin; draw v from Eq. (9) (rejection) and (θ, ϕ) from Eq. (11)
while any particle is in flight **do**
  Store $(\vec{r}, \vec{v})$ of all flying particles
  Advance all flying particles by Δt (one `solve_ivp` call)
  for each flying particle with $z < 0$ **do**
    $f \leftarrow z_{\text{old}}/(z_{\text{old}} - z_{\text{new}})$; interpolate $\vec{r}$ and $\vec{v}$ at f; set $z = 0$
    if $u < p_{\text{stick}}$ **then** stick
    else if $u' < p_s$ **then** $v_z \leftarrow -v_z$ ▷ specular
    else redraw direction, keep $|\vec{v}|$ ▷ diffuse
    end if
  end for
end while
Record final positions and bounce counts

The unit direction is $\hat{n} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta)$, and the initial velocity is $v\,\hat{n}$. Unlike rejection sampling, this method uses every random deviate [1]. The same distribution is used for diffuse reflections. The sampled distribution of θ is shown in Fig. 2.

IV. NUMERICAL METHOD

A. Algorithm

The simulation advances all particles in flight synchronously with a fixed outer time step Δt (Algorithm 1).

B. Time integration

The positions and velocities of all N_{fly} particles in flight are concatenated into a single state vector of length $6N_{\text{fly}}$ and advanced with `scipy.integrate.solve_ivp` [6], using the explicit fifth-order Dormand–Prince pair with embedded fourth-order error estimate (RK45) [4]. A single call therefore advances the whole ensemble. The initial step is set to Δt so that the solver attempts the full step at once; it subdivides only if the embedded error estimate exceeds the tolerance. Without drag the trajectory is quadratic in t, which RK45 integrates exactly up to round-off.

C. Landing detection

A floor crossing is detected at the end of a step, when $z_{\text{new}} < 0$. The crossing is located at the fraction $f = z_{\text{old}}/(z_{\text{old}} - z_{\text{new}})$ of the step, and both position and velocity are linearly interpolated to this fraction before the particle is placed on the floor. The error introduced by this procedure is analyzed in Sec. VI C.

D. Implementation

The code is distributed as an installable Python package, `bounce`, built on NumPy [5] and SciPy [6]. The command-line interface provides `bounce run` (single simulation), `bounce validate` (all checks of Sec. V and the error studies of Sec. VI), and `bounce benchmark` (timing). All modules are covered by unit tests written with `pytest`, and all runs use fixed random seeds, so every number and figure in this paper can be reproduced.

V. VALIDATION

We compare simulation output with analytically known results. Unless noted, the checks use $\Delta t = 0.01$, $\sigma = 1$, and $N = 10^4$; the range check uses 2×10^4, the energy check 2×10^3, and the terminal-velocity check 10^2 particles. For a sample mean the standard error is $s/\sqrt{N}$, with s the sample standard deviation, and we accept deviations within three standard errors. For the hypothesis tests we require $p \geq 0.01$ for the Kolmogorov–Smirnov test and $p \geq 0.003$, the 3σ level, for the χ^2 test. All checks pass (Table I).

a. Ballistic range. With $p_{\text{stick}} = 1$ and no drag each particle lands once, and its landing point must agree with Eq. (5). Since the crossing is resolved within one step, the error is bounded by the horizontal distance per step, $v_h \Delta t$; the worst particle attains 0.11 of this bound. Because speed and direction are independent, the mean range is $\langle v^2 \rangle \langle \sin 2\theta \rangle / g = 3\sigma^2 \cdot (\pi/4)/g = 2.356$, compared with the measured 2.363 ± 0.015.

b. Energy conservation. Without drag, energy can change only at a landing, through the interpolation. The chord between the two step endpoints lies below the parabolic trajectory, so at the interpolated crossing time the true particle is still above the floor, by at most

TABLE I. Validation checks ($\Delta t = 0.01$, $\sigma = 1$, $N = 10^4$). Tolerances for sample means are three standard errors.

Check	Simulation	Expected	Tolerance
Range: worst-case landing error/$(v_h \Delta t)$	0.11	0	≤ 1
Mean first-landing distance	2.363	2.356	± 0.045
Largest relative energy change, $\max \Delta E/E$	-3.2×10^{-8}	≤ 0 (no gain)	$\leq 10^{-12}$
Largest energy loss/$(n_{\text{land}}\, mg^2 \Delta t^2/8)$	0.999999	≤ 1	≤ 1
Mean final x	-0.028	0	± 0.068
Mean final y	0.022	0	± 0.069
Uniformity of ϕ (KS p-value)	0.57	—	≥ 0.01
Random walk: $\langle r^2/(k+1) \rangle$	9.95	10.0	± 0.73
Specular: $\langle r/((k+1)R) \rangle$	0.99987	1	± 0.001
Specular: change of direction	1.6×10^{-13}	0	$\leq 10^{-9}$
Terminal velocity: $\lvert v_z + v_t \rvert / v_t$	1.1×10^{-15}	0	$\leq 10^{-8}$
Mean number of bounces	3.967	4.000	± 0.133
Bounce distribution $P(k)$ (χ^2 p-value)	0.039	—	≥ 0.003

$g\Delta t^2/8$. The interpolation therefore can only remove energy, by at most $mg^2\Delta t^2/8$ per landing, and the simulation respects this bound.

This check exposed an error in an early version of the code, which used the end-of-step velocity at the landing. At that instant the particle has already fallen below the floor and has gained kinetic energy, so each bounce injected energy; some particles more than doubled their energy over their lifetime. Interpolating the velocity as well as the position removed the spurious energy gain entirely.

c. Random walk. With purely diffuse reflection ($p_s = 0$) the flight vectors of a particle are independent with zero mean, so after k bounces $\langle r^2 \rangle = (k+1)\langle R^2 \rangle$, with $\langle R^2 \rangle = \langle v^4 \rangle \langle \sin^2 2\theta \rangle / g^2 = 15\sigma^4 \cdot \frac{2}{3}/g^2 = 10\,\sigma^4/g^2$. We measure 9.95 ± 0.24.

d. Bounce statistics. The number of bounces must follow Eq. (6). For $p_{\text{stick}} = 0.2$ the expected mean is 4; we measure 3.97 ± 0.04. A χ^2 test of the full histogram against $P(k)$ gives $p = 0.039$, consistent with the geometric distribution at the chosen significance level.

e. Symmetry and limiting cases. Rotational symmetry of the source requires zero mean final x and y and a uniform azimuth; we test both with drag ($\gamma = 0.2$), the latter with a Kolmogorov–Smirnov test. With purely specular reflection all flights of a particle are identical, so it must come to rest at $r = (k+1)R$ along its initial azimuth. Finally, with $\gamma = 0.5$ a particle falling for a long time must approach $v_z = -mg/\gamma$.

VI. ERROR ANALYSIS

A. Round-off error

Since RK45 is exact for drag-free motion, the energy drift of a single flight without floor contact measures round-off alone. We integrate to $t = 2$ with an increasing number of steps (Table II). The error grows with the number of steps, as expected from accumulation of rounding errors, but remains below 10^{-10} and is negligible for all results reported here.

TABLE II. Maximum relative energy error due to round-off in drag-free flight to $t = 2$.

Steps	Δt	$\max \lvert \Delta E \rvert / E$
10	0.2	2.5×10^{-14}
100	0.02	1.4×10^{-13}
1000	0.002	1.0×10^{-13}
10000	0.0002	1.3×10^{-11}

TABLE III. Maximum position error of `solve_ivp` with drag ($\gamma = 0.5$, $t = 5$).

`rtol`	Single call	Steps of $\Delta t = 0.01$
10^{-3}	3.7×10^{-4}	1.2×10^{-14}
10^{-6}	3.9×10^{-7}	1.2×10^{-14}
10^{-10}	2.7×10^{-11}	1.2×10^{-14}

B. Integration error

We compare `solve_ivp` with Eqs. (3)–(4) for $\gamma = 0.5$ up to $t = 5$, with `atol` $= 10^{-3}$ `rtol` (Table III). When the solver integrates the whole interval in one call with adaptive steps, the error tracks `rtol`. In the simulation, however, the solver is restarted every $\Delta t = 0.01$; on such short intervals the local truncation error is far below any tolerance used, and the error is $\approx 10^{-14}$ independent of `rtol`.

TABLE IV. Mean per-landing errors from the simulation ($N = 10^4$) compared with Eq. (13) (units $m = g = 1$).

Δt	$\langle\lvert\Delta R\rvert\rangle$	$\pi\Delta t^2/24$	$\langle\Delta E\rangle$	$\Delta t^2/12$
0.04	2.03×10^{-4}	2.09×10^{-4}	1.33×10^{-4}	1.33×10^{-4}
0.02	5.06×10^{-5}	5.24×10^{-5}	3.34×10^{-5}	3.33×10^{-5}
0.01	1.27×10^{-5}	1.31×10^{-5}	8.33×10^{-6}	8.33×10^{-6}
0.005	3.16×10^{-6}	3.27×10^{-6}	2.06×10^{-6}	2.08×10^{-6}
0.0025	7.95×10^{-7}	8.18×10^{-7}	5.20×10^{-7}	5.21×10^{-7}

C. Time-step (interpolation) error

The dominant discretization error comes from the landing interpolation. Suppose the true crossing occurs at time $s \in [0, \Delta t]$ within the step. To leading order the interpolated crossing precedes the true one, and at that instant the particle is still at height

$$h(s) = \frac{g}{2}\, s(\Delta t - s) \tag{12}$$

above the floor. Assuming the crossing phase s is uniformly distributed, $\langle s(\Delta t - s)\rangle = \Delta t^2/6$, so $\langle h\rangle = g\Delta t^2/12$. The resulting energy deficit is $mg\,h$, and the horizontal error is $h\tan\theta$, where for drag-free flight the impact angle equals the launch angle. Using $\langle\tan\theta\rangle = \pi/2$ for $p(\theta) = \sin 2\theta$,

$$\langle\Delta E\rangle = \frac{mg^2\Delta t^2}{12}, \qquad \langle\lvert\Delta R\rvert\rangle = \frac{\pi g\Delta t^2}{24}. \tag{13}$$

Table IV compares these predictions with the simulation. The energy error agrees to within 1% and the range error to within 4%, and a power-law fit gives order $\Delta t^{2.00}$ for both.

D. Statistical error

All reported quantities are ensemble means, with standard error $s/\sqrt{N}$. For the mean final distance we obtain 0.30, 0.082, and 0.027 for $N = 10^3$, 10^4, and 10^5 (Table VIII); a log–log fit gives slope -0.53, consistent with $-1/2$ (Fig. 3). For the mean first-landing distance the exact value $3\pi\sigma^2/(4g)$ is known, so the actual error can also be measured. It is 0.083, 0.0044, and 0.0014 for the three sample sizes, of the same order as or smaller than the estimated standard error, as expected for a single realization.

E. Error budget

Table V summarizes the error sources at the production parameters. The statistical error exceeds all numerical errors by at least three orders of magnitude, so the accuracy of our results is limited by the sample size rather than by the numerics. Consequently, the most effective way to improve accuracy is to increase N, not to decrease Δt.

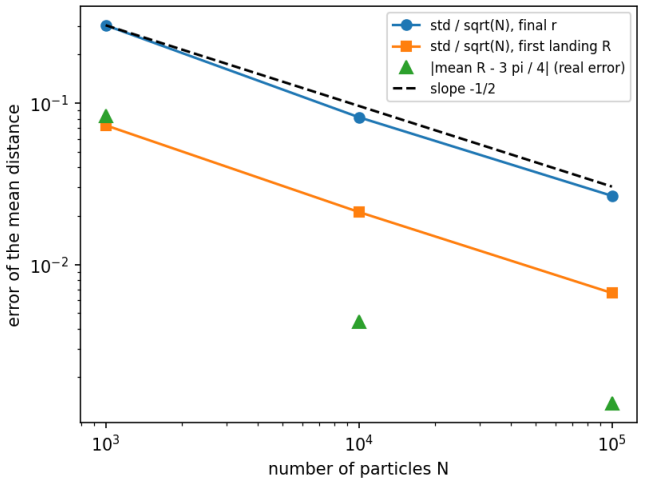


FIG. 3. Standard error $s/\sqrt{N}$ of the mean final distance and of the mean first-landing distance, together with the actual error $|\langle R\rangle - 3\pi/4|$ of the latter, as functions of N.

TABLE V. Error budget at $\Delta t = 0.01$ and $N = 10^4$.

Source	Scaling	Magnitude
Round-off	grows with steps	$< 10^{-10}$
Integration (RK45)	$\propto$ local truncation	$\sim 10^{-14}$
Landing interpolation	Δt^2	$\sim 10^{-5}$
Statistical sampling	$N^{-1/2}$	$\sim 10^{-2}$

VII. COMPUTATIONAL COST

A. Scaling analysis

In each outer step all $N_{\rm fly}$ flying particles are advanced. Each `solve_ivp` call performs one RK45 step with seven right-hand-side evaluations (we measure exactly 7.0 per call), each a vectorized operation on $6N_{\rm fly}$ numbers, while the surface routine touches only the particles that landed. The cost per step is therefore $O(N_{\rm fly})$ plus a fixed overhead per `solve_ivp` call.

The total cost is the number of particle-steps, $\sum_{\rm steps} N_{\rm fly}$. Each particle performs $k+1$ flights of mean duration $\langle 2v\cos\theta/g\rangle = 2\langle v\rangle\langle\cos\theta\rangle/g = 2\cdot 1.596\,\sigma\cdot\frac{2}{3}/g = 2.13\,\sigma/g$ (no drag), so the expected number of particle-steps per particle is

$$\frac{\langle k+1\rangle\cdot 2.13\,\sigma/g}{\Delta t} = \frac{2.13\,\sigma/g}{p_{\rm stick}\,\Delta t}, \tag{14}$$

and the total cost is $O(N/(p_{\rm stick}\,\Delta t))$. For $p_{\rm stick} = 0.2$ and $\Delta t = 0.01$, Eq. (14) gives 1064 particle-steps per

TABLE VI. Wall-clock time of one outer step with all particles in flight.

N	Time per step (s)	Time per particle (s)
10^3	1.7×10^{-4}	1.7×10^{-7}
10^4	1.1×10^{-3}	1.1×10^{-7}
10^5	2.1×10^{-2}	2.1×10^{-7}

particle. The number of outer steps is set by the longest-lived particle. Since the maximum of N geometric variates grows as $\ln N / \ln[1/(1-p_{\mathrm{stick}})]$, the number of outer steps grows only logarithmically in N, although with large fluctuations from run to run. Memory usage is $O(N)$.

B. Measured scaling

The wall-clock time of a single step with all particles in flight scales as $N^{1.04}$ (Table VI), confirming linear per-step cost. The measured particle-steps per particle are 1156, 1057, and 1069 for $N = 10^3$, 10^4, and 10^5 (Table VIII), in agreement with the prediction of 1064; the larger value at $N = 10^3$ reflects the sampling fluctuation in $\langle k \rangle$ discussed in Sec. VIII A. Varying the time step at $N = 10^4$ (Table IX) yields 272, 1057, and 4276 particle-steps per particle, compared with the predicted 266, 1064, and 4256, confirming the $1/\Delta t$ scaling.

C. Benchmarks and profiling

Complete runs required 1.6, 5.6, and 36 s for $N = 10^3$, 10^4, and 10^5, an effective scaling of $N^{0.68}$. The sublinear scaling arises because most outer steps contain only a few remaining particles, so the fixed per-call overhead of `solve_ivp` dominates at small N. Accordingly, the time per particle-step decreases from 1.4×10^{-6} s to 5.3×10^{-7} s and 3.4×10^{-7} s, approaching a constant only when the particle work dominates.

A `cProfile` analysis of an $N = 10^4$ run (5.4 s total) is summarized in Table VII. The flight routine dominates, and within it most time is spent in the solver's internal Runge–Kutta machinery rather than in the physics: the right-hand side accounts for only 7%. Counting the flying particles each step (`n_flying`) also costs 7%, more than anticipated. Replacing `solve_ivp` with the exact propagator, Eqs. (3)–(4), makes the same run approximately three times faster.

Benchmarking also revealed excessive memory use: for $N = 10^5$ the resident memory grew to several gigabytes because solver objects from previous calls were not released promptly. Explicitly releasing them after each `solve_ivp` call reduced peak memory to about 170 MB, consistent with the expected $O(N)$ footprint.

TABLE VII. Cumulative-time profile of a run with $N = 10^4$.

Function	Fraction of run time
`move` (flight, total)	74%
`solve_ivp`	62%
Runge–Kutta step	47%
Right-hand side	7%
`n_flying`	7%
`land` (surface interaction)	$< 4\%$

TABLE VIII. Results for three ensemble sizes ($\Delta t = 0.01$). PS/N denotes particle-steps per particle.

N	$\langle r \rangle$	$\langle k \rangle$	max k	Steps	PS/N
10^3	7.23 ± 0.30	4.41 ± 0.16	43	13799	1156
10^4	6.54 ± 0.08	3.97 ± 0.04	43	10923	1057
10^5	6.59 ± 0.03	4.007 ± 0.014	62	24039	1069

VIII. RESULTS

Unless stated otherwise, we use $p_{\mathrm{stick}} = 0.2$, $p_s = 0.5$, $\sigma = 1$, and no drag. The ensemble and time-step studies use seed 0, and the single run shown in Figs. 4–6 uses seed 1.

A. Convergence with the number of particles

Table VIII lists results for $N = 10^3$, 10^4, and 10^5. The uncertainties of the mean final distance and mean bounce number decrease as $N^{-1/2}$. At $N = 10^5$ the mean bounce number is 4.007 ± 0.014, in agreement with the exact value of 4. At $N = 10^3$ it is 4.41 ± 0.16, a 2.6σ deviation, illustrating that small ensembles can deviate substantially from the expectation by chance. The maximum bounce number increases with N, as larger ensembles sample further into the tail of $P(k)$. The mean first-landing distance converges to $3\pi\sigma^2/(4g) = 2.356$: we obtain 2.44 ± 0.07, 2.361 ± 0.021, and 2.355 ± 0.007.

B. Convergence with the time step

Table IX shows results for three time steps at $N = 10^4$. Variations of the mean final distance are within the statistical uncertainty, consistent with the error budget of Table V: the $O(\Delta t^2)$ interpolation error is negligible, while the cost grows as $1/\Delta t$. For this problem, $\Delta t = 0.04$ would therefore give equally accurate results at a quarter of the cost.

TABLE IX. Results for three time steps ($N = 10^4$).

Δt	$\langle r \rangle$	$\langle k \rangle$	PS/N
0.04	6.63 ± 0.09	4.06	272
0.01	6.54 ± 0.08	3.97	1057
0.0025	6.66 ± 0.08	4.02	4276

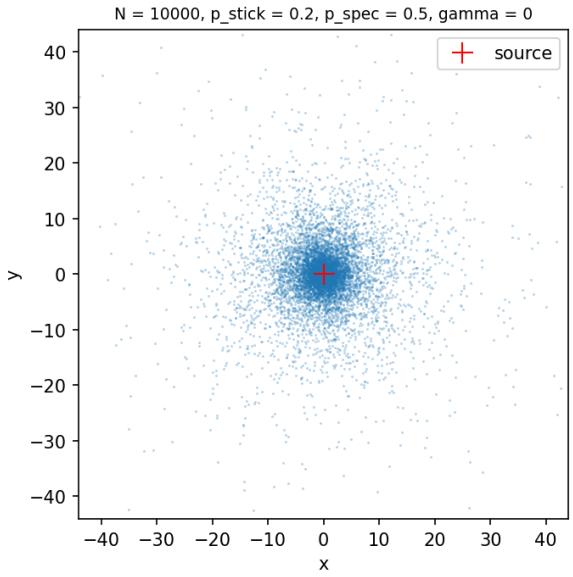


FIG. 4. Final positions of $N = 10^4$ particles on the floor.

C. Spatial distribution and bounce statistics

Figure 4 shows the final positions from a single run with $N = 10^4$. The deposit is azimuthally symmetric about the source, as required by the rotational symmetry of the emission. The radial distribution (Fig. 5) is strongly peaked near the source with a long tail produced by particles with many bounces. In this run the mean final distance is 6.79 and the mean first-landing distance is 2.365 (exact: 2.356). The bounce-number histogram (Fig. 6) is linear on a logarithmic scale, as expected for the geometric distribution of Eq. (6), with mean 4.03.

D. Specular versus diffuse reflection and the effect of drag

Figure 7 compares purely specular and purely diffuse reflection, with and without drag ($N = 10^4$). Under specular reflection and no drag, a particle retains its direction and its displacements add coherently, giving a mean final distance of 12.1. Under diffuse reflection the displacements form a random walk, with $\langle r^2 \rangle$ growing only linearly in the number of flights, and the mean final distance is 4.5. Linear drag with $\gamma = 0.3$ shortens every flight, reducing the mean distances to 2.6 (specular) and

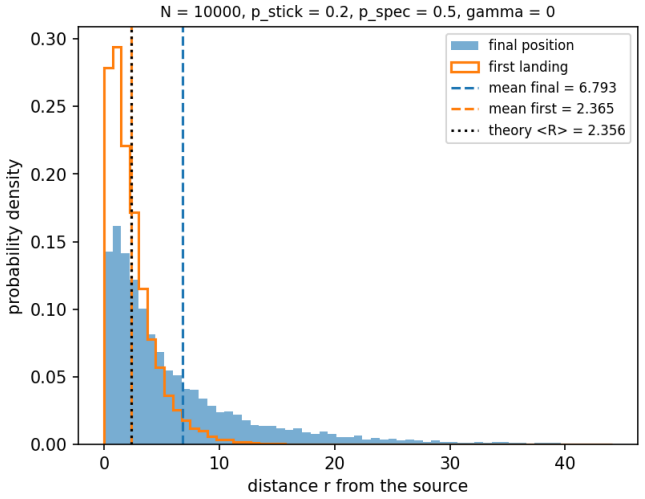


FIG. 5. Distributions of the final distance r and of the first-landing distance ($N = 10^4$). The dotted line marks $\langle R \rangle = 3\pi\sigma^2/(4g)$.

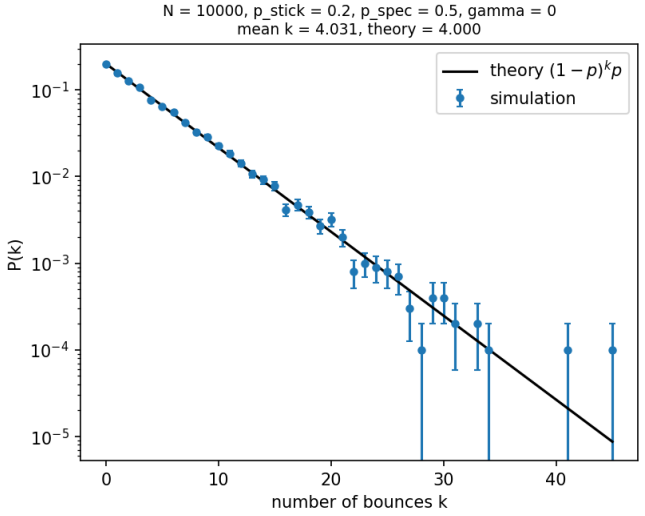


FIG. 6. Bounce-number distribution ($N = 10^4$) compared with $P(k)$, Eq. (6). Error bars are $\sqrt{\text{count}}/N$.

1.7 (diffuse).

IX. DISCUSSION AND CONCLUSIONS

We have developed and validated a Monte Carlo simulation of particles undergoing ballistic flight with drag and stochastic surface interactions. All thirteen validation checks pass, and the energy check was instrumental in identifying an interpolation error in an early version of the code. The error analysis shows a clear hierarchy: round-off ($< 10^{-10}$) and integration ($\sim 10^{-14}$) errors are negligible, the landing-interpolation error scales as Δt^2 with an analytically predicted prefactor confirmed to

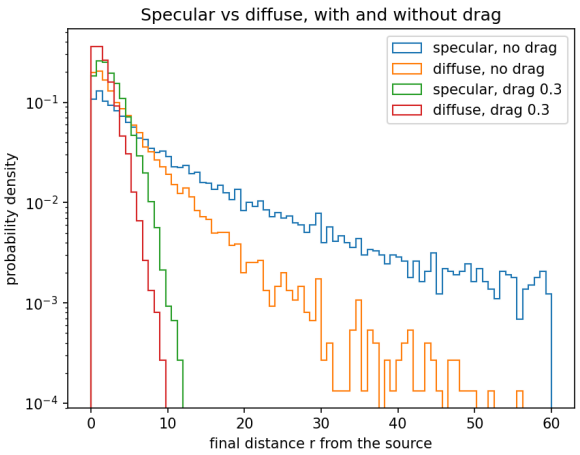


FIG. 7. Final-distance distributions for purely specular and purely diffuse reflection, with and without drag ($\gamma = 0.3$, $N = 10^4$).

within a few percent, and the statistical error ($\propto N^{-1/2}$) dominates by more than three orders of magnitude at the production parameters. The computational cost scales as $O(N/(p_{\mathrm{stick}}\,\Delta t))$, in agreement with measurement, with `solve_ivp` accounting for most of the run time.

Several limitations suggest directions for future work. The model assumes elastic reflection; a coefficient of restitution would introduce energy dissipation at the surface and change the long-distance tail. Linear drag is appropriate only at low Reynolds number; quadratic drag would remove the closed-form solution but could still be validated against the drag-free limit. On the numerical side, because the free-flight equation is linear, the exact propagator of Eqs. (3)–(4) combined with root-finding for the crossing time would eliminate both the interpolation error and most of the run time. Finally, since the results are sampling-limited and the error budget permits a larger Δt, the available computational effort is best spent on larger ensembles.

CODE AVAILABILITY

The `bounce` package, including the tests, the validation and benchmark commands, and the script `make_figures.py` that produces all figures, is available at https://github.com/JavadYousefian/Project-2--Computational-Physics (tag `v1.0`).

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